Academic Presentation

Precision-Aware Adaptive Incremental SVD

Bounding Orthogonality Drift in Streaming Subspaces

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The Bottleneck of Batch SVD

Goal: Isolate moving objects via optimal low-rank matrix extraction.

The Math: For a video matrix $X \in \mathbb{R}^{p \times q}$ (where $q \to \infty$).

The Problem: Batch SVD requires $O(pq \min(p,q))$ operations.

Conclusion: Batch SVD is computationally impossible for continuous streams.

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The Baseline: Deferred SVD (Brand, 2006)

The 5-Factor Representation: $X \approx U\, U'\, \Sigma\, V'^{T}\, V^{T}$

U (Tall Basis): Never rotated, grows via cheap column appends.

U' (Small Core): Absorbs all plane rotations in $O(r^3)$.

Result: Strictly linear-time $O(pqr)$ processing.

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The Fatal Vulnerability

Finite-Precision Reality:
$fl(a \circ b) = (a \circ b)(1 + \delta)$

Consequence:
Cumulative basis orthogonality drift ($||U^TU - I_r||F > 0$)

Brand (2006):
"It is an open question how often [re-orthogonalization] is necessary to guarantee numerical precision."

The Dilemma:
Naive Gram-Schmidt repair costs $O(pr^2)$, destroying linear-time speed.

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Our Solution: PA-ISVD

Exact O(pr) Drift Tracking: No explicit Gram matrix.

Wilkinson's Adaptive Trigger: Mathematical precision bounds.

Lossless Subspace Absorption: Safe basis repair.

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Contribution 1: Exact O(pr) Drift Tracking

Theorem (Recursive Frobenius Drift):

$$\|U_{new}^T U_{new} - I_{r+1}\|_F^2 = d_U^2 + 2\|U^T P\|_2^2 + \left(\|P\|_2^2 - 1\right)^2$$

Standard Gram Check: O(pr2) flops.

Our Recursive Check: strictly O(pr) flops.

Note: Scalar tracker runs in float64 to prevent meta-drift.

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Contribution 2: Wilkinson's Adaptive Trigger

Wilkinson's Backward Error Bound: Inner product roundoff ≈ $r \cdot u$.

Adaptive Threshold: $\tau = k \cdot r \cdot u$ (where u = unit roundoff)

Two-Tier Repair Hierarchy:

- Tier 1: Core refactor in $O(r^3)$ flops.
- Tier 2: Tall MGS in $O(pr^2)$ flops (Invoked strictly if $dU > \tau$).

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Contribution 3: Lossless Subspace Absorption

Lemma (Subspace Absorption Invariance):
$UU'\Sigma = (QR)U'\Sigma = Q(RU')\Sigma$

Step 1: $U \leftarrow Q$ (Restored to machine precision).
Step 2: Fold R into the core: $M = RU'$.

Step 3: Compute SVD of $M\Sigma$.

Result: Spatial basis repaired, but data matrix X remains algebraically identical.

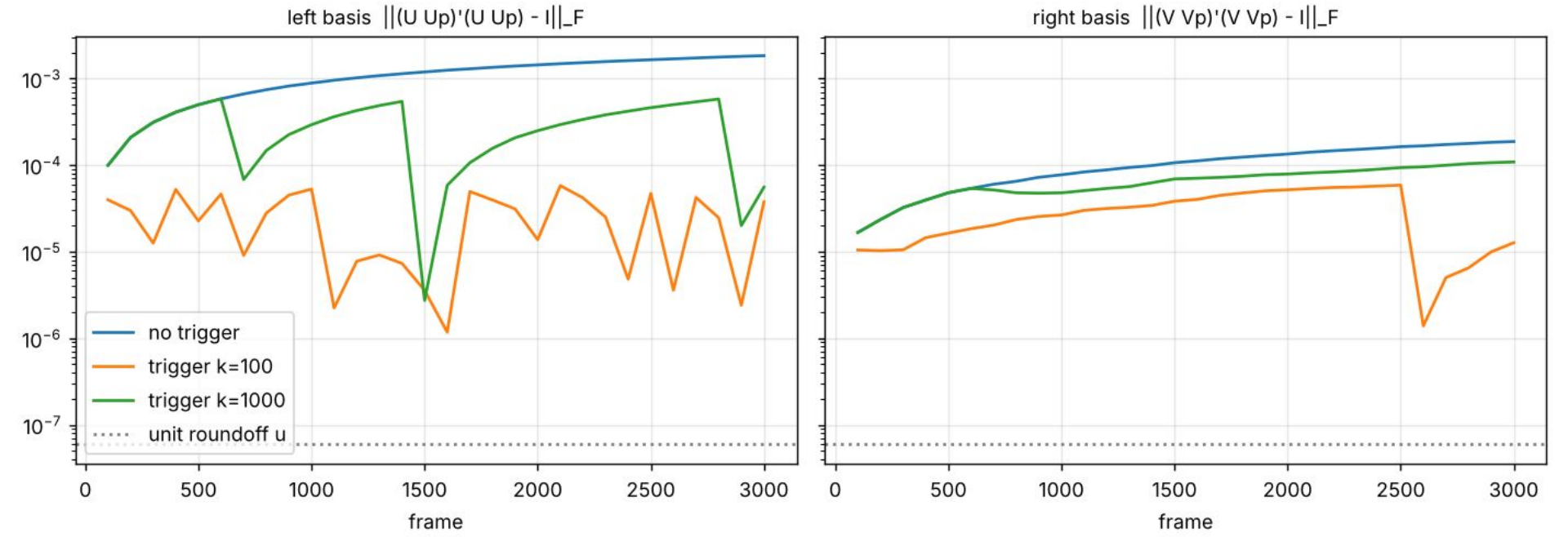
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Empirical Validation: Bounding Orthogonality Drift



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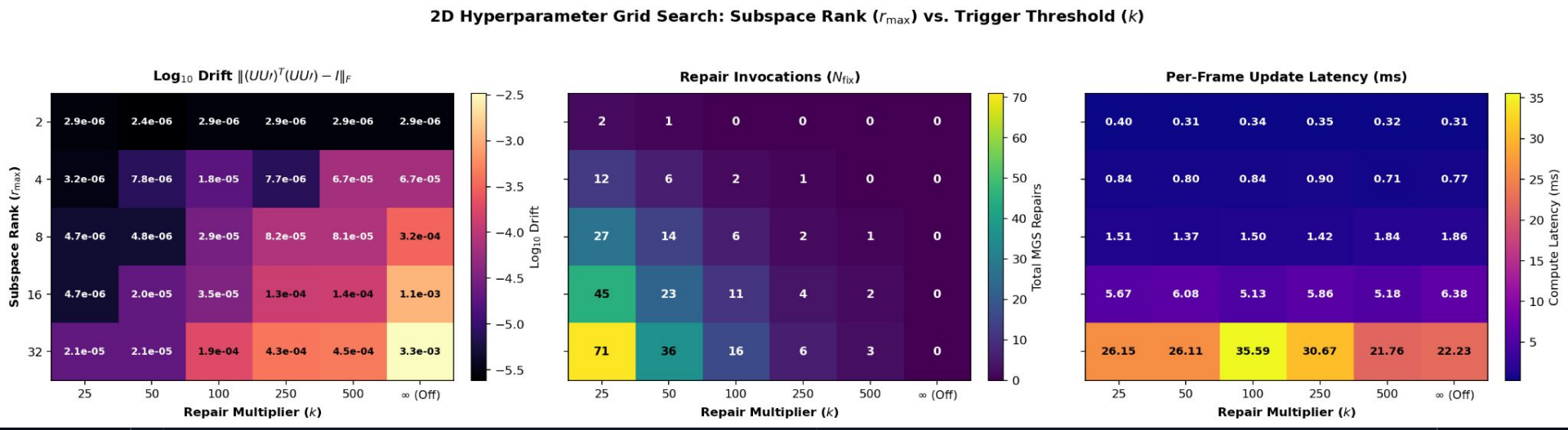
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2D Pareto Hyperparameter Grid Search

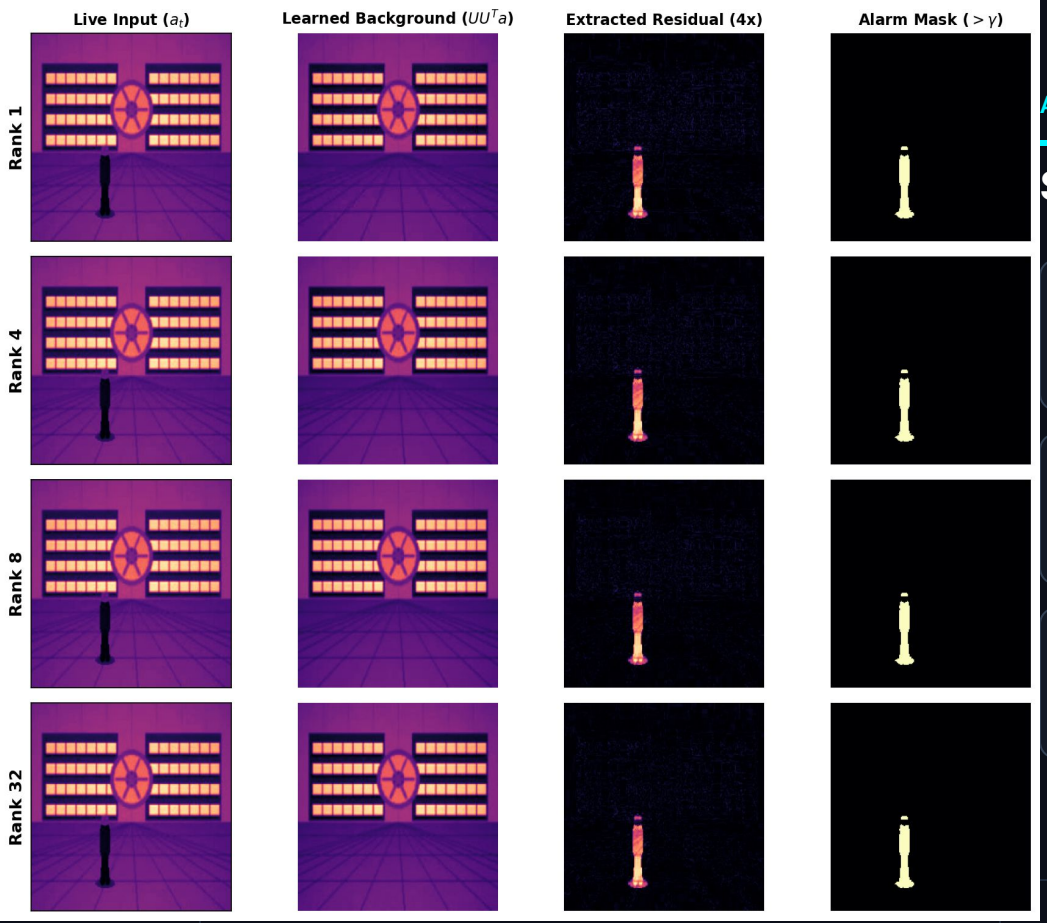


- Evaluated 30 configurations (r × k).
- Left: Drift bounded to ≤ 10^{-5} across all ranks.
- Center: Repair frequency scales inversely with k.
- Right: Update latency is dominated by r; repairs add zero observable overhead.

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Subspace Capacity Ablation

Rank 1: 4.4× higher numerical clutter floor.

Rank 32: Subspace leakage (overfitting), 24ms latency penalty.

Rank 8: Pareto Optimal (4.7× SCR, 1.86ms latency).

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Real-World Deployment: Vault CCTV



- Dimensionality: p = 49,152 pixels.
- Throughput: > 500 FPS (sub-2ms latency).
- Hardware Efficiency: Entire core matrix (< 4 KB) resides in CPU L1-Cache.

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Conclusion

Resolved Brand's (2006) Open Problem with mathematically guaranteed numerical stability.

Novel Contributions:

- O(pr) drift tracker
- Wilkinson adaptive trigger
- Lossless subspace absorption

Impact:

Infinite-horizon streaming PCA with strict O(pqr) complexity guarantees.

Code:

`github.com/yashdighade774/CLA-Project`

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